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(54) **CIRCUIT AND METHOD FOR BIASING AN AMPLIFIER**

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H03F 1/22 (2006.01)
H03F 3/45 (2006.01)

(52) **U.S. Cl.**

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(2013.01); **H03F 2200/451** (2013.01)

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3/30
USPC **330/311**, **296**, **285**
See application file for complete search history.

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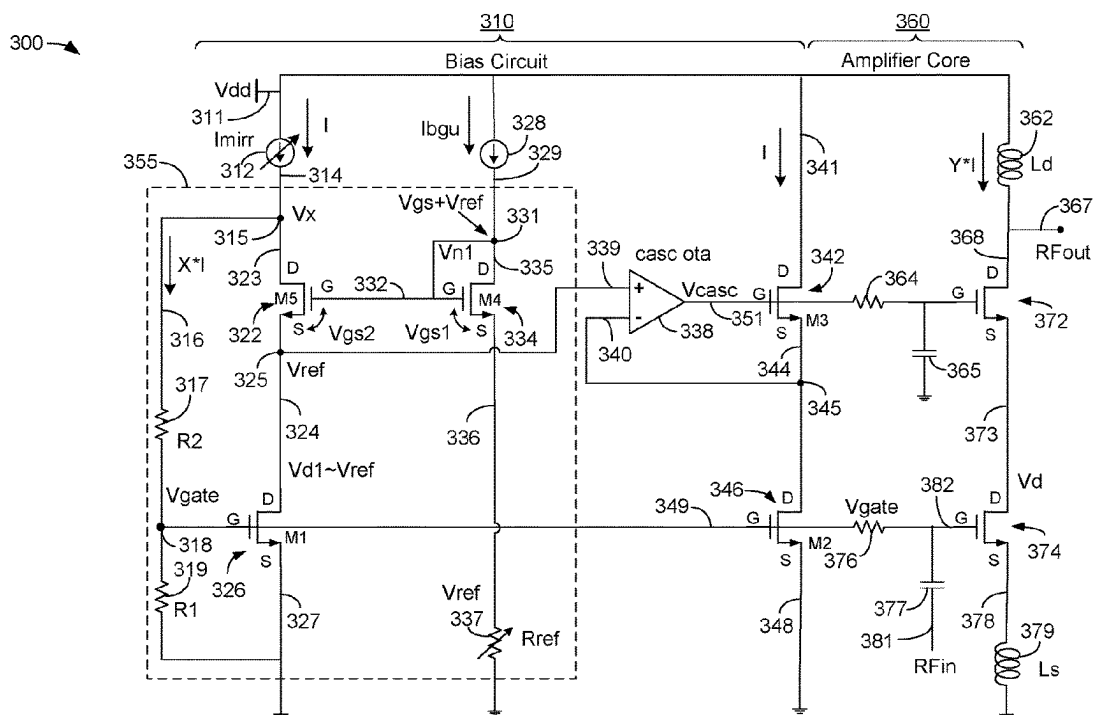
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(57) **ABSTRACT**

An amplifier circuit includes an amplifier core having a cascode transistor and a gain transistor, a bias circuit coupled to the amplifier core, the bias circuit comprising: a first current source, a second current source, an operational transconductance amplifier (OTA), a bias cascode transistor pair having a bias cascode transistor and a bias gain transistor, and a replica circuit coupled to the first current source and to the second current source.

14 Claims, 7 Drawing Sheets



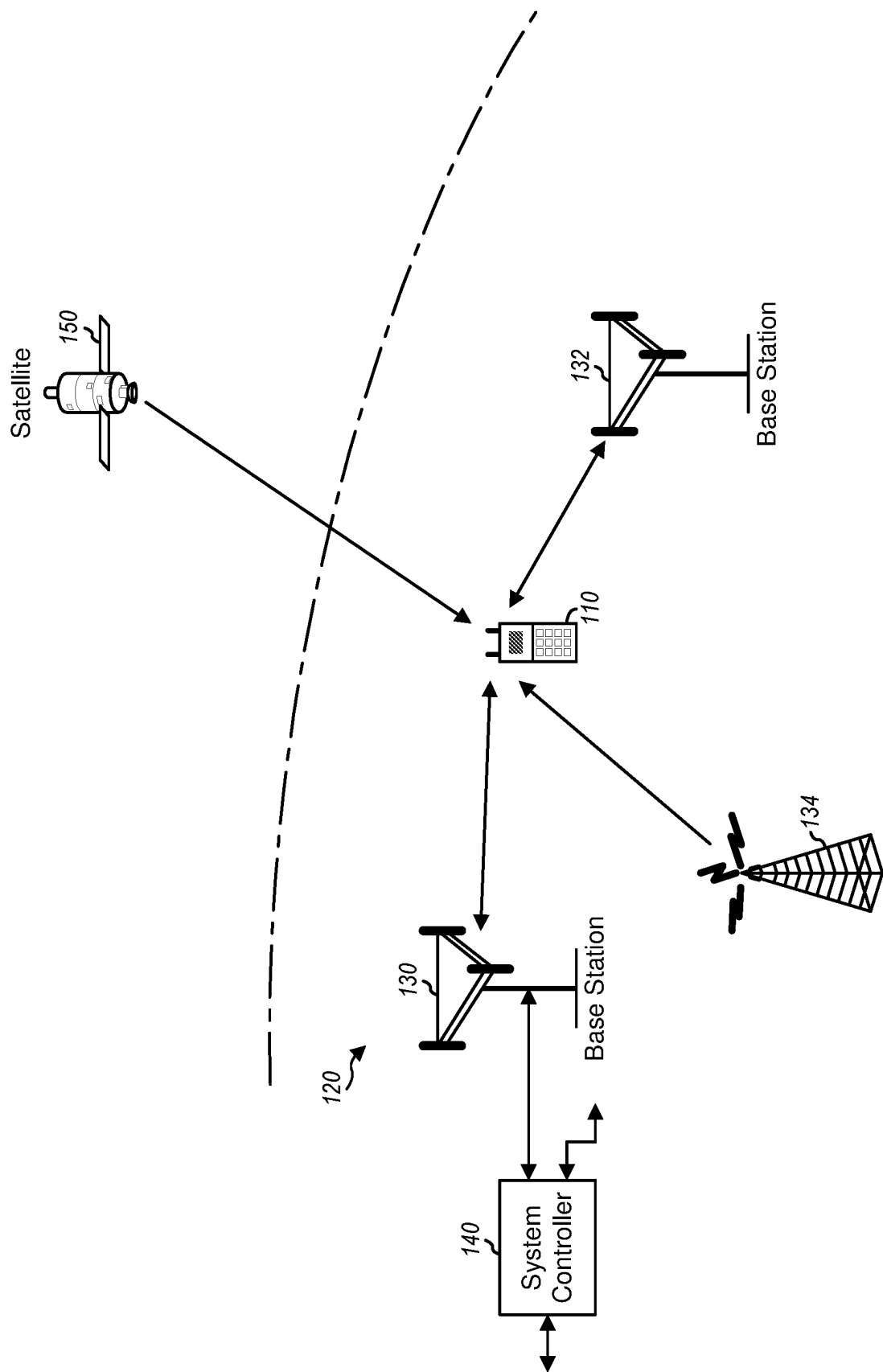


FIG. 1

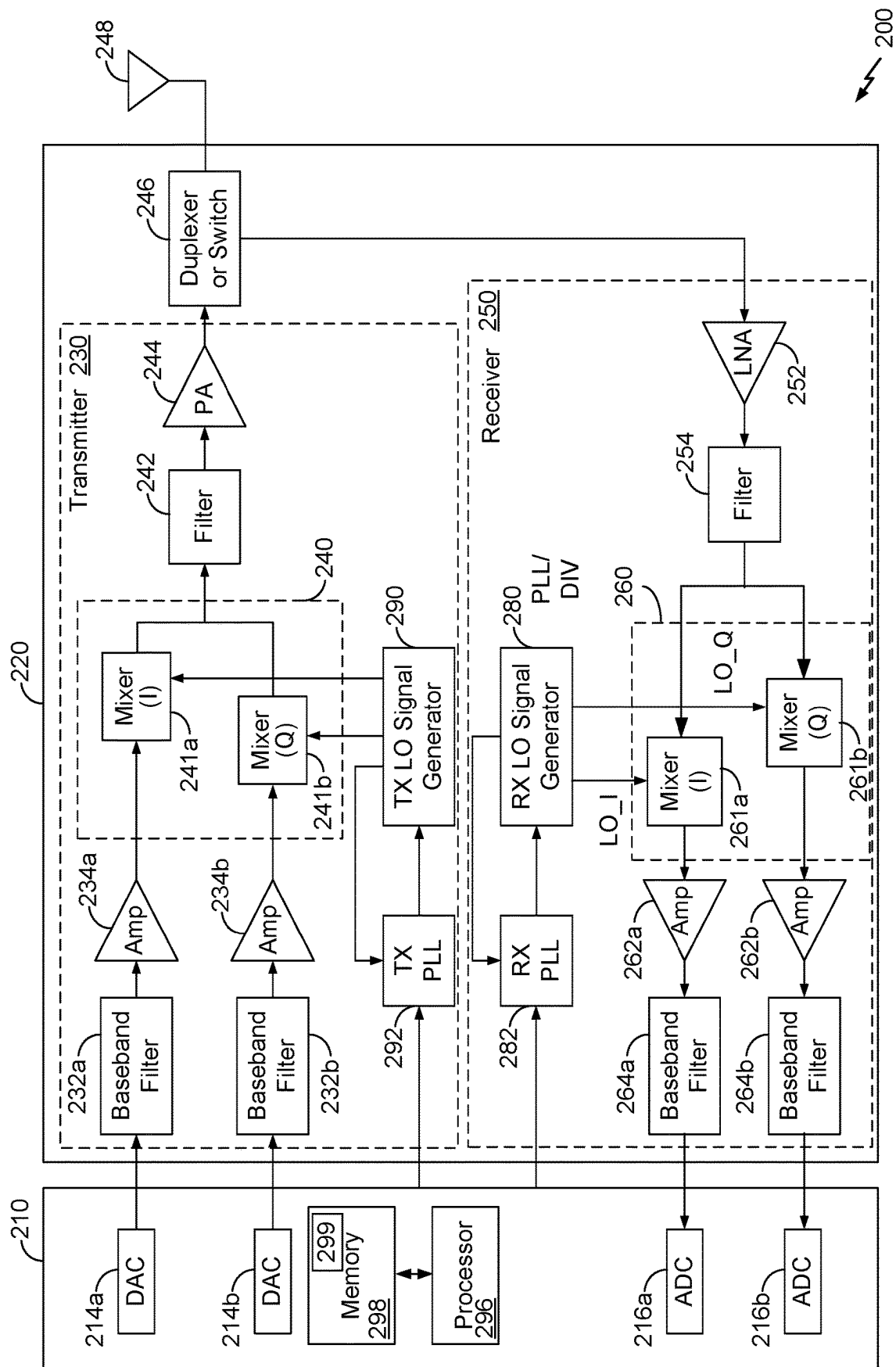


FIG. 2A

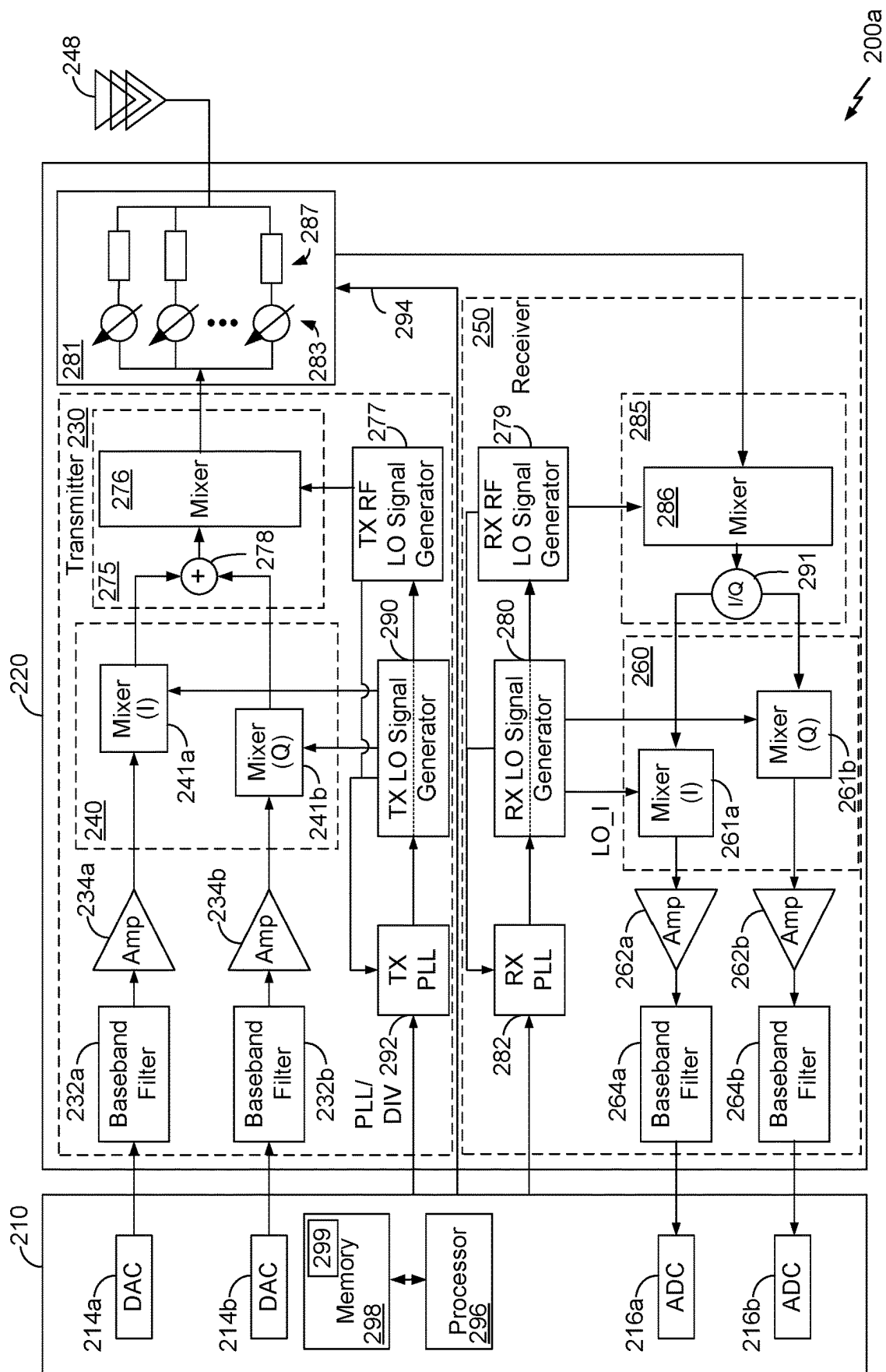


FIG. 2B

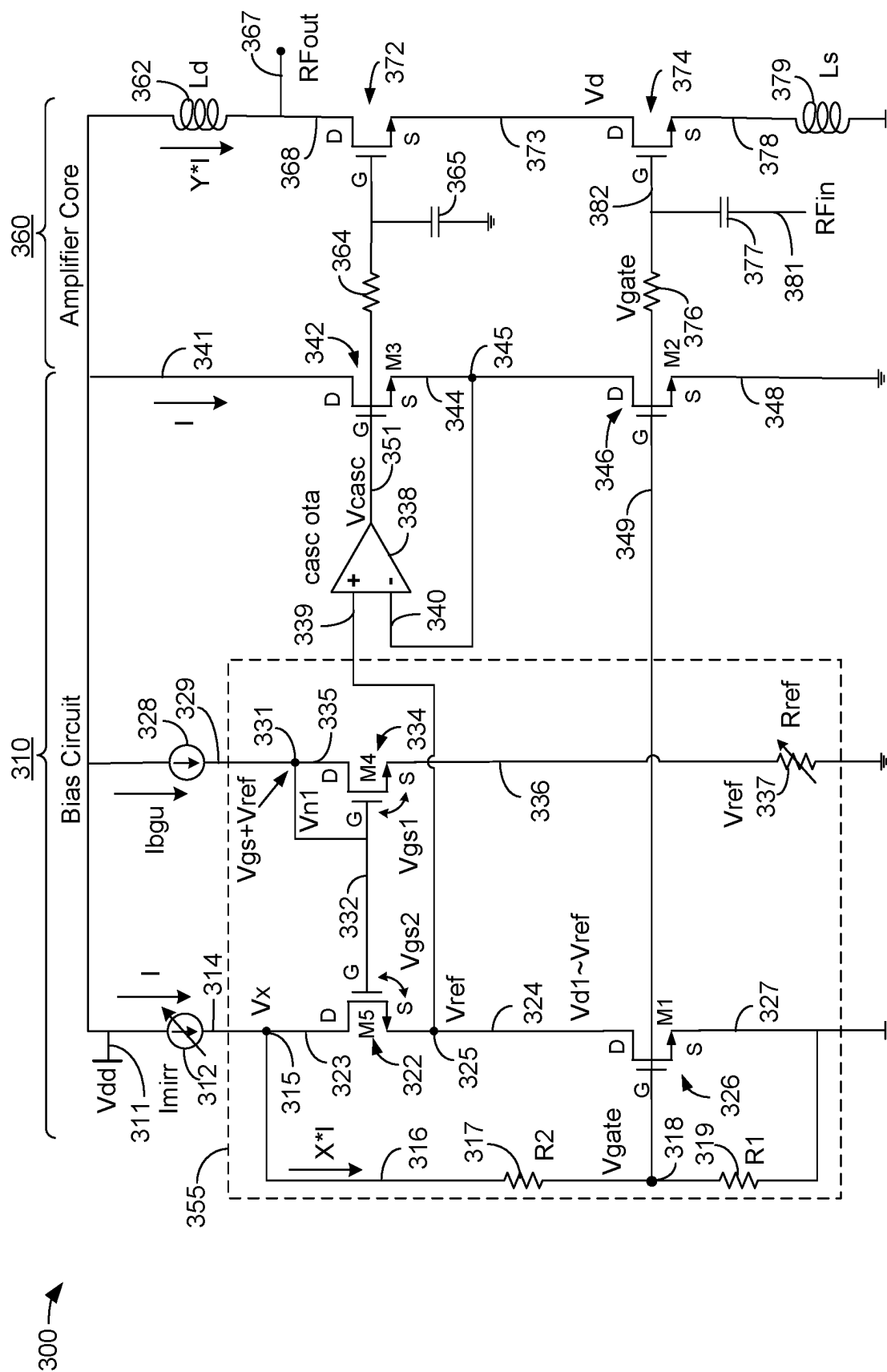
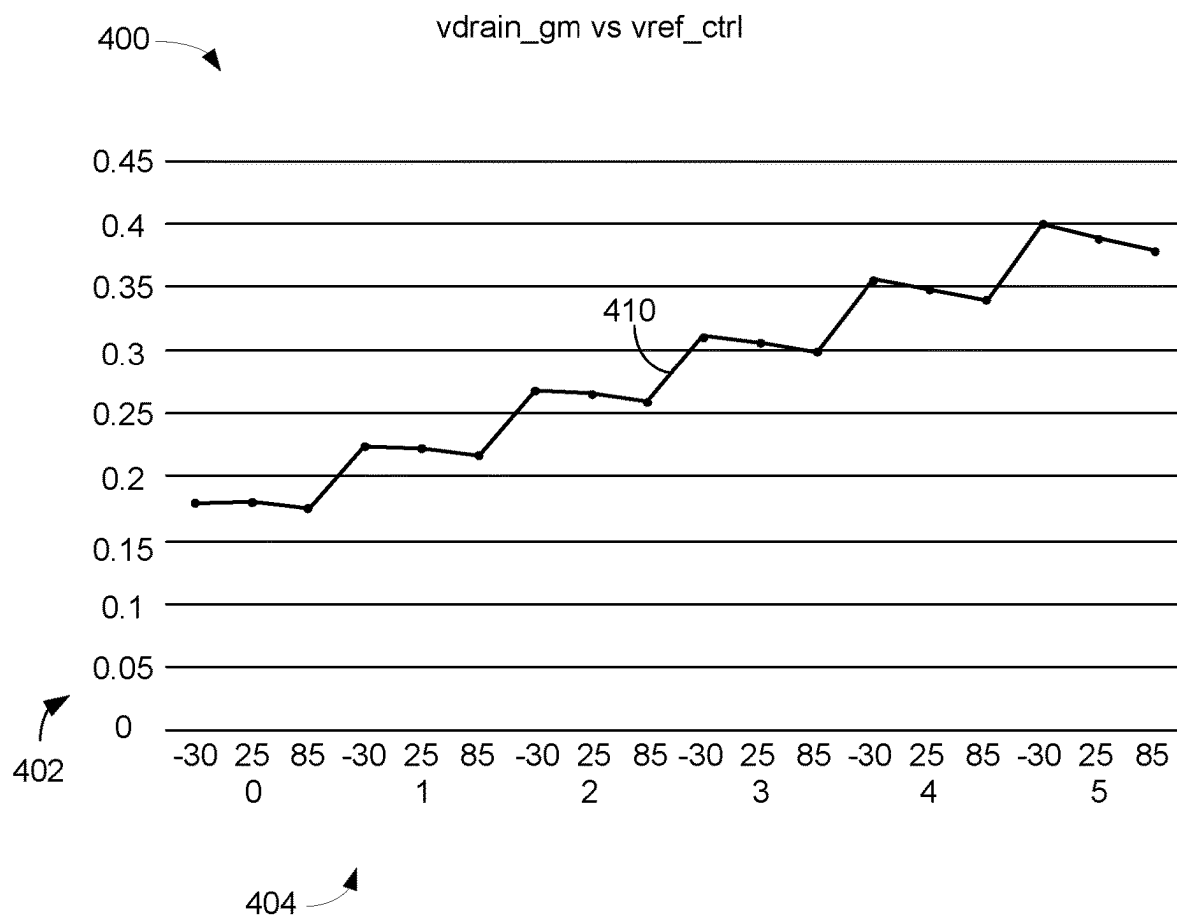
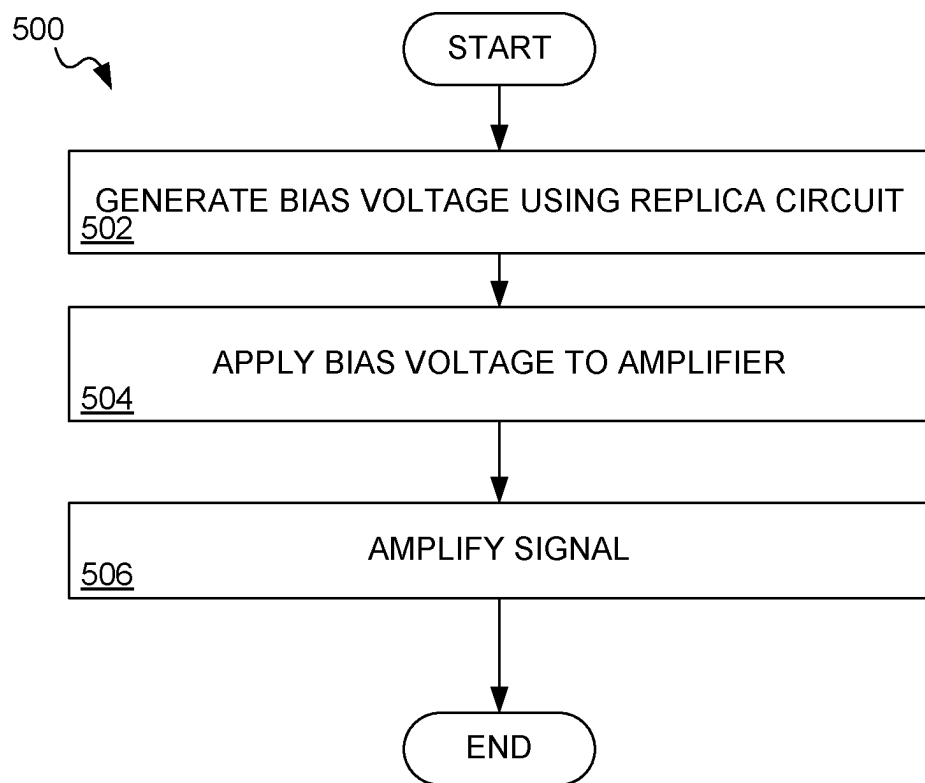
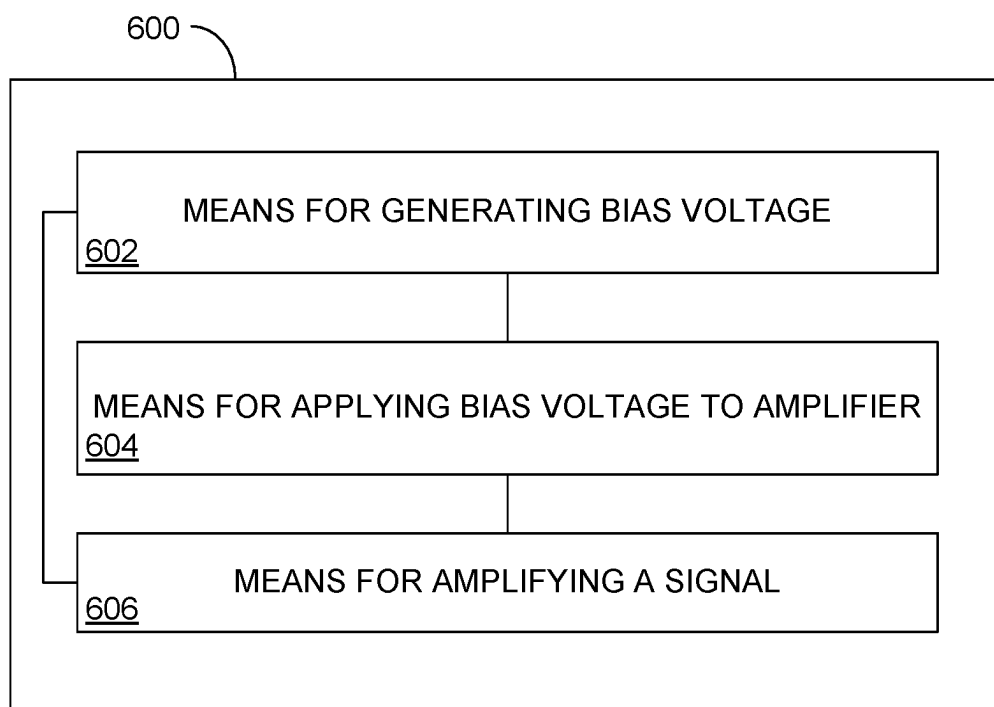


FIG. 3

**FIG. 4**

**FIG. 5**

**FIG. 6**

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CIRCUIT AND METHOD FOR BIASING AN AMPLIFIER

RELATED APPLICATIONS

This application claims priority to and the benefit of U.S. Provisional Patent Application No. 63/271,966, entitled “CIRCUIT AND METHOD FOR BIASING AN AMPLIFIER,” filed Oct. 26, 2021, the contents of which are hereby incorporated herein by reference in their entirety as if fully set forth below and for all applicable purposes.

FIELD

The present disclosure relates generally to electronics, and more specifically to bias circuits for amplifiers.

BACKGROUND

Wireless communication devices and technologies are becoming ever more prevalent, as are communication devices that operate at millimeter-wave (mmW) frequencies. Wireless communication devices generally transmit and/or receive communication signals. In a radio frequency (RF) transceiver, a communication signal is typically amplified and transmitted by a transmit section and a received communication signal is amplified and processed by a receive section. A receive section may comprise one or more circuits that amplify and process the communication signal. The amplifier circuit or circuits may comprise one or more amplifier paths having one or more stages that may include one or more low noise amplifier (LNA) circuits. One of the factors that determines how the LNA operates is a bias circuit. A bias circuit typically establishes the operating parameters of an amplifier. Accordingly, it is desirable to operate the bias circuit as efficiently as possible.

SUMMARY

Various implementations of systems, methods and devices within the scope of the appended claims each have several aspects, no single one of which is solely responsible for the desirable attributes described herein. Without limiting the scope of the appended claims, some prominent features are described herein.

Details of one or more implementations of the subject matter described in this specification are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages will become apparent from the description, the drawings, and the claims. Note that the relative dimensions of the following figures may not be drawn to scale.

One aspect of the disclosure provides an amplifier circuit including an amplifier core having a cascode transistor and a gain transistor, a bias circuit coupled to the amplifier core, the bias circuit having a first current source, a second current source, an operational transconductance amplifier (OTA), a bias cascode transistor pair having a bias cascode transistor and a bias gain transistor and a replica circuit coupled to the first current source and to the second current source. The replica includes a first replica transistor coupled to the first current source, and a second replica transistor coupled to the second current source, a third replica transistor coupled to the second replica transistor, an adjustable resistance coupled to the first replica transistor, a voltage divider coupled to the second replica transistor and the third replica

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transistor, where a gate of the third replica transistor provides a bias voltage, V_{gate} , for the amplifier core.

Another aspect of the disclosure provides a method for biasing an amplifier, including generating a bias voltage using a first current loop and a second current loop, the first current loop comprising a plurality of current sources and a voltage divider, the second current loop comprising a single amplifier, and applying the bias voltage to an amplifier.

Another aspect of the disclosure provides a device including means for generating a bias voltage using a first current loop and a second current loop, the first current loop comprising a plurality of current sources and a voltage divider, the second current loop comprising a single amplifier, and means for applying the bias voltage to an amplifier.

BRIEF DESCRIPTION OF THE DRAWINGS

In the figures, like reference numerals refer to like parts throughout the various views unless otherwise indicated. For reference numerals with letter character designations such as “102a” or “102b”, the letter character designations may differentiate two like parts or elements present in the same figure. Letter character designations for reference numerals may be omitted when it is intended that a reference numeral encompass all parts having the same reference numeral in all figures.

FIG. 1 is a diagram showing a wireless device communicating with a wireless communication system.

FIG. 2A is a block diagram showing a wireless device in which exemplary techniques of the present disclosure may be implemented.

FIG. 2B is a block diagram showing a wireless device in which exemplary techniques of the present disclosure may be implemented.

FIG. 3 is a block diagram of an amplifier circuit in accordance with an exemplary embodiment of the disclosure.

FIG. 4 is a graph having a trace showing the drain voltage, V_d , of an LNA gain transistor as a function of V_{ref} and temperature.

FIG. 5 is a flow chart describing an example of the operation of a method for biasing an amplifier.

FIG. 6 is a functional block diagram of an apparatus for biasing an amplifier.

DETAILED DESCRIPTION

The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects.

In a communication device receiver, a low noise amplifier (LNA) may be biased by a circuit that includes transistors, which are typically implemented using a process technology referred to as metal-oxide semiconductor field effect transistor (MOSFET). A MOSFET may also be referred to as a FET and may also be a p-channel FET (PFET) or an n-channel FET (NFET). A complementary metal-oxide semiconductor (CMOS) circuit may include both NFETs and PFETs. Regardless of the implementation type, a PET device typically provides a gain which may also be referred to as the G_m of the device. The measurement of the linearity of a FET device used as a low noise amplifier (LNA) and the noise figure, or NF, of the device may be important characteristics.

A FET typically has three terminals, referred to as a gate, a source and a drain. Various voltages exist in a FET, a

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gate-source voltage is referred to as V_{gs} ; a drain-source voltage is referred to as V_{ds} ; a drain voltage is referred to as V_d or V_{drain} ; and a gate voltage is referred to as V_g or V_{gate} .

An LNA using FETs may be implemented as a transconductance amplifier (also referred to as a G_m stage) using what is referred to as a cascode architecture, where the cascode architecture includes two transistors, one transistor operating as a common source and the other transistor operating as a common gate. One of the transistors (the gain transistor) receives the RF input signal at its gate, and the other transistor (the cascode transistor) provides the RF signal output at its drain.

In a low noise amplifier (LNA) that is implemented using FET devices the LNA bias current is dependent on the drain voltage, V_d , of the amplifier device that is providing the signal amplification (the G_m , or gain) of the LNA, that is, the gain transistor. Such a device may be fabricated having a short channel length, which may reduce the NF or increase the likelihood that a NF requirement is satisfied. The gate-source voltage, V_{gs} and the drain-source voltage, V_{ds} , should be set correctly to control the bias current, the linearity, and/or the noise figure (NF) of the gain transistor of the LNA. The dependence on the drain voltage, V_d , of the gain transistor that is providing the signal amplification (the G_m , or gain) of the LNA becomes more significant at low bias current, which is typically used for lower gain modes (such as for the G4 mode, which is typically about 1 mA (milliAmp)).

Existing bias circuits use two current loops to bias the LNA. This provides good control over the LNA bias current, particularly with process, voltage and temperature (PVT) variations, and allows the gain (G_m) of the gain transistor to be set independently of the drain voltage, V_d . However, such a bias circuit may be implemented using two operational transconductance amplifiers (OTAs), which results in significant power consumption (particularly in lower gain modes), and consumes significant area on the circuit.

Therefore, it would be desirable to have a way of efficiently biasing an amplifier.

In an exemplary embodiment, a bias circuit for a low noise amplifier (LNA) provides the above-mentioned dual loop behavior, while using only a single OTA.

In an exemplary embodiment, a bias circuit for a low noise amplifier (LNA) may be implemented using a smaller circuit area and may consume less power due to being implemented using a single OTA.

FIG. 1 is a diagram showing a wireless device **110** communicating with a wireless communication system **120**. The wireless communication system **120** may be a Long Term Evolution (LTE) system, a Code Division Multiple Access (CDMA) system, a Global System for Mobile Communications (GSM) system, a wireless local area network (WLAN) system, a 5G NR (new radio) system, or some other wireless system. A CDMA system may implement Wideband CDMA (WCDMA), CDMA 1×, Evolution-Data Optimized (EVDO), Time Division Synchronous CDMA (TD-SCDMA), or some other version of CDMA. For simplicity, FIG. 1 shows wireless communication system **120** including two base stations **130** and **132** and one system controller **140**. In general, a wireless communication system may include any number of base stations and any set of network entities.

The wireless device **110** may also be referred to as a user equipment (UE), a mobile station, a terminal, an access terminal, a subscriber unit, a station, etc. Wireless device **110** may be a cellular phone, a smartphone, a tablet, a

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wireless modem, a personal digital assistant (PDA), a handheld device, a laptop computer, a smartbook, a netbook, a tablet, a cordless phone, a medical device, an automobile, a device configured to connect to one or more other devices (for example through the internet of things), a wireless local loop (WLL) station, a Bluetooth device, etc. Wireless device **110** may communicate with wireless communication system **120**. Wireless device **110** may also receive signals from broadcast stations (e.g., a broadcast station **134**) and/or signals from satellites (e.g., a satellite **150** in one or more global navigation satellite systems (GNSS)), etc). Wireless device **110** may support one or more radio technologies for wireless communication such as LTE, WCDMA, CDMA 1×, EVDO, TD-SCDMA, GSM, 802.11, 802.15, 5G, UWB, etc.

Wireless device **110** may support carrier aggregation, for example as described in one or more LTE or 5G standards. In some embodiments, a single stream of data is transmitted over multiple carriers using carrier aggregation, for example as opposed to separate carriers being used for respective data streams. Wireless device **110** may be able to operate in a variety of communication bands including, for example, those communication bands used by LTE, WiFi, 5G or other communication bands, over a wide range of frequencies. Wireless device **110** may also be capable of communicating directly with other wireless devices without communicating through a network.

In general, carrier aggregation (CA) may be categorized into two types—*intra-band CA* and *inter-band CA*. *Intra-band CA* refers to operation on multiple carriers within the same band. *Inter-band CA* refers to operation on multiple carriers in different bands.

FIG. 2A is a block diagram showing a wireless device **200** in which exemplary techniques of the present disclosure may be implemented. The wireless device **200** may, for example, be an embodiment of the wireless device **110** illustrated in FIG. 1.

FIG. 2A shows an example of a transceiver **220** having a transmitter **230** and a receiver **250**. In general, the conditioning of the signals in the transmitter **230** and the receiver **250** may be performed by one or more stages of amplifier, filter, upconverter, downconverter, etc. These circuit blocks may be arranged differently from the configuration shown in FIG. 2A. Furthermore, other circuit blocks not shown in FIG. 2A may also be used to condition the signals in the transmitter **230** and receiver **250**. Unless otherwise noted, any signal in FIG. 2A, or any other figure in the drawings, may be either single-ended or differential. Some circuit blocks in FIG. 2A may also be omitted.

In the example shown in FIG. 2A, wireless device **200** generally comprises the transceiver **220** and a data processor **210**. The data processor **210** may include a processor **296** operatively coupled to a memory **298**. The memory **298** may be configured to store data and program codes shown generally using reference numeral **299**, and may generally comprise analog and/or digital processing components. The processor **296** and the memory **298** may cooperate to control, configure, program, or otherwise fully or partially control some or all of the operation of the embodiments of the system and method for biasing an amplifier described herein.

The transceiver **220** includes a transmitter **230** and a receiver **250** that support bi-directional communication. In general, wireless device **200** may include any number of transmitters and/or receivers for any number of communication systems and frequency bands. All or a portion of the

transceiver **220** may be implemented on one or more analog integrated circuits (ICs), RF ICs (RFICs), mixed-signal ICs, etc.

A transmitter or a receiver may be implemented with a super-heterodyne architecture or a direct-conversion architecture. In the super-heterodyne architecture, a signal is frequency-converted between radio frequency (RF) and baseband in multiple stages, e.g., from RF to an intermediate frequency (IF) in one stage, and then from IF to baseband in another stage for a receiver. In the direct-conversion architecture, a signal is frequency converted between RF and baseband in one stage. The super-heterodyne and direct-conversion architectures may use different circuit blocks and/or have different requirements. In the example shown in FIG. 2A, transmitter **230** and receiver **250** are implemented with the direct-conversion architecture.

In the transmit path, the data processor **210** processes data to be transmitted and provides in-phase (I) and quadrature (Q) analog output signals to the transmitter **230**. In an exemplary embodiment, the data processor **210** includes digital-to-analog-converters (DAC's) **214a** and **214b** for converting digital signals generated by the data processor **210** into the I and Q analog output signals, e.g., I and Q output currents, for further processing. In other embodiments, the DACs **214a** and **214b** are included in the transceiver **220** and the data processor **210** provides data (e.g., for I and Q) to the transceiver **220** digitally.

Within the transmitter **230**, baseband (e.g., lowpass) filters **232a** and **232b** filter the I and Q analog transmit signals, respectively, to remove undesired images caused by the prior digital-to-analog conversion. Amplifiers (Amp) **234a** and **234b** amplify the signals from baseband filters **232a** and **232b**, respectively, and provide I and Q baseband signals. An upconverter **240** having upconversion mixers **241a** and **241b** upconverts the I and Q baseband signals with I and Q transmit (TX) local oscillator (LO) signals from a TX LO signal generator **290** and provides an upconverted signal. A filter **242** filters the upconverted signal to remove undesired images caused by the frequency upconversion as well as noise in a receive frequency band. A power amplifier (PA) **244** amplifies the signal from filter **242** to obtain the desired output power level and provides a transmit RF signal. The transmit RF signal may be routed through a duplexer or switch **246** and transmitted via an antenna **248**. While examples discussed herein utilize I and Q signals, those of skill in the art will understand that components of the transceiver may be configured to utilize polar modulation.

In the receive path, antenna **248** receives communication signals and provides a received RF signal, which may be routed through duplexer or switch **246** and provided to a low noise amplifier (LNA) **252**. The duplexer **246** is designed to operate with a specific RX-to-TX duplexer frequency separation, such that RX signals are isolated from TX signals. The received RF signal is amplified by LNA **252** and filtered by a filter **254** to obtain a desired RF input signal. Exemplary embodiments of the system and method for biasing an amplifier may be implemented in or as part of the LNA **252**. In other embodiments, the system and method for biasing an amplifier may be implemented in other portions of the receiver **250**, or in a transceiver, such as in a transceiver circuit, module or chip.

Downconversion mixers **261a** and **261b** in a downconverter **260** mix the output of filter **254** with I and Q receive (RX) LO signals (i.e., LO_I and LO_Q) from an RX LO signal generator **280** to generate I and Q baseband signals. The I and Q baseband signals are amplified by amplifiers **262a** and **262b** and further filtered by baseband (e.g., low-

pass) filters **264a** and **264b** to obtain I and Q analog input signals, which are provided to data processor **210**. In the exemplary embodiment shown, the data processor **210** includes analog-to-digital-converters (ADC's) **216a** and **216b** for converting the analog input signals into digital signals to be further processed by the data processor **210**. In some embodiments, the ADCs **216a** and **216b** are included in the transceiver **220** and provide data to the data processor **210** digitally.

In FIG. 2A, TX LO signal generator **290** generates the I and Q TX LO signals used for frequency upconversion, while RX LO signal generator **280** generates the I and Q RX LO signals used for frequency downconversion. Each LO signal is a periodic signal with a particular fundamental frequency. A phase locked loop (PLL) **292** receives timing information from data processor **210** and generates a control signal used to adjust the frequency and/or phase of the TX LO signals from LO signal generator **290**. Similarly, a PLL **282** receives timing information from data processor **210** and generates a control signal used to adjust the frequency and/or phase of the RX LO signals from LO signal generator **280**.

Wireless device **200** may support CA and may (i) receive multiple downlink signals transmitted by one or more cells on multiple downlink carriers at different frequencies and/or (ii) transmit multiple uplink signals to one or more cells on multiple uplink carriers. Those of skill in the art will understand, however, that aspects described herein may be implemented in systems, devices, and/or architectures that do not support carrier aggregation.

Certain components of the transceiver **220** are functionally illustrated in FIG. 2A, and the configuration illustrated therein may or may not be representative of a physical device configuration in certain implementations. For example, as described above, transceiver **220** may be implemented in various integrated circuits (ICs), RF ICs (RFICs), mixed-signal ICs, etc. In some embodiments, the transceiver **220** is implemented on a substrate or board such as a printed circuit board (PCB) having various modules, chips, and/or components. For example, the power amplifier **244**, the filter **242**, and the duplexer **246** may be implemented in separate modules or as discrete components, while the remaining components illustrated in the transceiver **220** may be implemented in a single transceiver chip.

The power amplifier **244** may comprise one or more stages comprising, for example, driver stages, power amplifier stages, or other components, that can be configured to amplify a communication signal on one or more frequencies, in one or more frequency bands, and at one or more power levels. Depending on various factors, the power amplifier **244** can be configured to operate using one or more driver stages, one or more power amplifier stages, one or more impedance matching networks, and can be configured to provide good linearity, efficiency, or a combination of good linearity and efficiency.

In an exemplary embodiment in a super-heterodyne architecture, the PA **244** and LNA **252** (and filter **242** and filter **254** in some examples) may be implemented separately from other components in the transmitter **230** and receiver **250**, for example on a millimeter wave integrated circuit. An example super-heterodyne architecture is illustrated in FIG. 2B.

FIG. 2B is a block diagram showing a wireless device in which exemplary techniques of the present disclosure may be implemented. Certain components, for example which may be indicated by identical reference numerals, of the wireless device **200a** in FIG. 2B may be configured similarly

to those in the wireless device **200** shown in FIG. 2A and the description of identically numbered items in FIG. 2B will not be repeated.

The wireless device **200a** is an example of a heterodyne (or superheterodyne) architecture in which the upconverter **240** and the downconverter **260** are configured to process a communication signal between baseband and an intermediate frequency (IF). For example, the upconverter **240** may be configured to provide an IF signal to an upconverter **275**. In an exemplary embodiment, the upconverter **275** may comprise summing function **278** and upconversion mixer **276**. The summing function **278** combines the I and the Q outputs of the upconverter **240** and provides a non-quadrature signal to the mixer **276**. The non-quadrature signal may be single ended or differential. The mixer **276** is configured to receive the IF signal from the upconverter **240** and TX RF LO signals from a TX RF LO signal generator **277**, and provide an upconverted RF signal to phase shift circuitry **281**. While PLL **292** is illustrated in FIG. 2B as being shared by the signal generators **290**, **277**, a respective PLL for each signal generator may be implemented.

In an exemplary embodiment, components in the phase shift circuitry **281** may comprise one or more adjustable or variable phased array elements, and may receive one or more control signals from the data processor **210** over connection **294** and operate the adjustable or variable phased array elements based on the received control signals.

In an exemplary embodiment, the phase shift circuitry **281** comprises phase shifters **283** and phased array elements **287**. Although three phase shifters **283** and three phased array elements **287** are shown for ease of illustration, the phase shift circuitry **281** may comprise more or fewer phase shifters **283** and phased array elements **287**.

Each phase shifter **283** may be configured to receive the RF transmit signal from the upconverter **275**, alter the phase by an amount, and provide the RF signal to a respective phased array element **287**. Each phased array element **287** may comprise transmit and receive circuitry including one or more filters, amplifiers, driver amplifiers, and/or power amplifiers. In some embodiments, the phase shifters **283** may be incorporated within respective phased array elements **287**.

The output of the phase shift circuitry **281** is provided to an antenna array **248**. In an exemplary embodiment, the antenna array **248** comprises a number of antennas that typically correspond to the number of phase shifters **283** and phased array elements **287**, for example such that each antenna element is coupled to a respective phased array element **287**. In an exemplary embodiment, the phase shift circuitry **281** and the antenna array **248** may be referred to as a phased array.

In a receive direction, an output of the phase shift circuitry **281** is provided to a downconverter **285**. In an exemplary embodiment, the downconverter **285** may comprise an I/Q generation function **291** and a downconversion mixer **286**. In an exemplary embodiment, the mixer **286** downconverts the receive RF signal provided by the phase shift circuitry **281** to an IF signal according to RX RF LO signals provided by an RX RF LO signal generator **279**. The I/Q generation function **291** receives the IF signal from the mixer **286** and generates I and Q signals for the downconverter **260**, which downconverts the IF signals to baseband, as described above. While PLL **282** is illustrated in FIG. 2B as being shared by the signal generators **280**, **279**, a respective PLL for each signal generator may be implemented.

In some embodiments, the upconverter **275**, downconverter **285**, and the phase shift circuitry **281** are implemented

on a common IC. In some embodiments, the summing function **278** and the I/Q generation function **291** are implemented separate from the mixers **276** and **286** such that the mixers **276**, **286** and the phase shift circuitry **281** are implemented on the common IC, but the summing function **278** and I/Q generation function **291** are not (e.g., the summing function **278** and I/Q generation function **291** are implemented in another IC coupled to the IC having the mixers **276**, **286**). In some embodiments, the LO signal generators **277**, **279** are included in the common IC. In some embodiments in which phase shift circuitry is implemented on a common IC with **276**, **286**, **277**, **278**, **279**, and/or **291**, the common IC and the antenna array **248** are included in a module, which may be coupled to other components of the transceiver **220** via a connector. In some embodiments, the phase shift circuitry **281**, for example, a chip on which the phase shift circuitry **281** is implemented, is coupled to the antenna array **248** by an interconnect. For example, components of the antenna array **248** may be implemented on a substrate and coupled to an integrated circuit implementing the phase shift circuitry **281** via a flexible printed circuit.

In some embodiments, both the architecture illustrated in FIG. 2A and the architecture illustrated in FIG. 2B are implemented in the same device. For example, a wireless device **110** or **200** may be configured to communicate with signals having a frequency below about 20 GHz using the architecture illustrated in FIG. 2A and to communicate with signals having a frequency above about 20 GHz using the architecture illustrated in FIG. 2B. In devices in which both architectures are implemented, one or more components of FIGS. 2A and 2B that are identically numbered may be shared between the two architectures. For example, both signals that have been downconverted directly to baseband from RF and signals that have been downconverted from RF to baseband via an IF stage may be filtered by the same baseband filter **264**. In other embodiments, a first version of the filter **264** is included in the portion of the device which implements the architecture of FIG. 2A and a second version of the filter **264** is included in the portion of the device which implements the architecture of FIG. 2B. While certain example frequencies are described herein, other implementations are possible. For example, signals having a frequency above about 20 GHz (e.g., having a mmW frequency) may be transmitted and/or received using a direct conversion architecture. In such embodiments, for example, a phased array may be implemented in the direct conversion architecture.

FIG. 3 is a block diagram of an amplifier circuit **300** in accordance with an exemplary embodiment of the disclosure. In an exemplary embodiment, the amplifier circuit **300** includes a bias circuit **310** and an amplifier core **360**. In some examples, the amplifier circuit **300** is configured as a low noise amplifier (LNA) and the amplifier core **360** comprises an LNA core. For example, the amplifier **300** may be an example of the LNA **252**, or of an LNA in one or more of the phased array elements **287**. In some such examples, the LNA is included in a front end module coupled to a transceiver of the wireless device **110** and/or includes short channel length devices. In other examples, the amplifier **300** may be configured as another type of amplifier, for example any of the other amplifiers described above and/or illustrated in FIG. 2, and may be included in a transceiver, in a front end module, in an RFIC, as a discrete component on a circuit board, etc.

In an exemplary embodiment, the bias circuit **310** includes a current source **312**, a current source **328** and a connection **341**, all coupled to a system voltage, V_{dd}, at a

node **311**. In an exemplary embodiment, the system voltage, V_{dd} , may be approximately 1.2V, or may be another voltage. In an exemplary embodiment, the current source **312** may be an adjustable or variable current source configured to provide an adjustable current across different gain modes. In an exemplary embodiment, the current source **312** may receive a control signal from the data processor **210** of FIG. 2A or FIG. 2B.

In an exemplary embodiment, the bias circuit **310** also includes a transistor **322**, a transistor **334**, a transistor **326**, an operational transconductance amplifier (OTA) **338**, a transistor **342**, a transistor **346**, a voltage divider (illustrated herein as a resistive voltage divider including resistance **317** and resistance **319**) and a variable resistance **337**. In an exemplary embodiment, the variable resistance **337** may receive a control signal from the data processor **210** of FIG. 2A or FIG. 2B.

The current source **312** is coupled to a node **315** over connection **314**. A drain of the transistor **322** is coupled to the node **315** over connection **323**. A source of the transistor **322** is coupled to a node **325**, and a gate of the transistor **322** is coupled to a gate of the transistor **334** over connection **332**. The node **315** may be coupled to an input of the voltage divider. While the voltage divider is configured as a resistive voltage divider in the example illustrated in FIG. 3, other types of voltage dividers may be used.

The current source **328** is coupled to a node **331** over connection **329**. A drain of the transistor **334** is coupled to the node **331** over connection **335**. A source of the transistor **334** is coupled to the variable resistance **337** over connection **336**. A gate of the transistor **334** and the gate of the transistor **322** are also coupled to the node **331**.

A drain of the transistor **326** is coupled to the node **325** over connection **324**. A source of the transistor **326** is coupled to system ground over connection **327**. A gate of the transistor **326** is coupled to a node **318**, which may comprise or be coupled to an output of the voltage divider. In the resistive voltage divider illustrated in FIG. 3, the resistance **317** is coupled between the node **315** and the node **318**. The resistance **319** is coupled between the node **318** and system ground.

A drain of the transistor **342** is coupled to the system voltage, V_{dd} , over connection **341**, a source of the transistor **342** is coupled to a node **345** and a gate of the transistor **342** coupled to an output of the OTA **338** over connection **351**.

A drain of the transistor **346** is coupled to the source of the transistor **342** over connection **347** at the node **345**, a source of the transistor **346** is coupled to system ground over connection **348**, and a gate of the transistor **346** is coupled to the node **318** over connection **349**.

The OTA **338** has a non-inverting input coupled to the node **325** over connection **339** and an inverting input coupled to the node **345** over connection **340**. A voltage at the node **325** may be referred to as V_{d1} (the voltage at the drain of the transistor **326**), which also closely approximates a reference voltage, V_{ref} , at the non-inverting input to the OTA **338**.

A gate-source voltage, V_{gs1} , refers to the gate-source voltage of the transistor **334** and a gate-source voltage, V_{gs2} , refers to the gate-source voltage of the transistor **322**. A voltage equal to $V_{gs} + V_{ref}$ appears at the node **331**. The voltage, V_{gs} , at node **331** refers to the gate-source voltage of the transistor **334** and the voltage, V_{n1} , is the sum of V_{gs} of the transistor **334** and the voltage, V_{ref} , across the variable resistance **337**. A voltage, V_x , appears at the node **315**.

The amplifier core **360** includes an optional inductance **362**, a transistor **372**, a transistor **374**, an optional inductance **379**, a resistance **364**, a capacitance **365**, a resistance **376** and a capacitance **377**. One side of the inductance **362** is coupled to the system voltage, V_{dd} , at node **311**. A drain of the transistor **372** is coupled to the other side of the inductance **362** over connection **368**. A source of the transistor **372** is coupled to a drain of the transistor **374** over connection **373**, and a gate of the transistor **372** is coupled to the resistance **364**. The other side of the resistance **364** is coupled to the gate of the transistor **342** and to the output of the OTA **338**. The gate of the transistor **372** is also coupled to the capacitance **365**.

A source of the transistor **374** is coupled to one side of the inductance **379** over connection **378**. The other side of the inductance **379** is coupled to system ground. The inductance **379** may be referred to as a source degeneration inductance. A gate of the transistor **374** is coupled to the resistance **376** and to the capacitance **377**. The other side of the resistance **376** is coupled to the gate of the transistor **346** and also to the node **318** over connection **349**. A radio frequency (RF) input signal (RF_{in}) is provided to the amplifier core **360** at connection **381** and an RF output signal, RF_{out} , is provided from the drain of the transistor **372** over connection **367**.

The transistor **374** may be fabricated as a short channel length device having a close dependency between gate voltage and drain voltage.

The arrangement of the transistors **342** and **346**, and the transistors **372** and **374** is referred to as a cascode arrangement. The transistors **342** and **372** are the cascode transistors and the transistors **346** and **374** are the gain transistors.

In an exemplary embodiment, the OTA **338** generates a voltage, V_{casc} , which is provided to the gate of the transistor **342** and to the gate of the transistor **372** through the resistance **364**.

In an exemplary embodiment, the transistors **326**, **346** and **342** may have a size referred to as (W/L) . The transistor **334** has a size that is referred to as $(W1/L1)$, the transistor **322** has a size that is referred to as $N1 \cdot (W1/L1)$, and the transistors **372** and **374** have a size that is referred to as $N \cdot (W/L)$.

The transistors **322**, **326**, **334**, **342**, **346**, **372** and **374** are depicted as N-type metal oxide semiconductor (NMOS) transistors; however, they may also be implemented using P-type devices.

In an exemplary embodiment, the current source **328** generates and flows a current, I_{bgu} , and the current source **312** generates and flows a current that mirrors the current I_{bgu} , and is referred to as a current, I_{mirror} . A current, I , flows through the connection **314** and through the connection **341**. The current, I , through the current source **312** is used to bias the transistor **374** and is typically higher than the current, I_{bgu} , generated by the current source **328**.

In an exemplary embodiment, the transistor **322**, the transistor **334**, the transistor **326**, the voltage divider (e.g., the resistance **317** and the resistance **319**) and the variable resistance **337** may comprise a replica circuit **355** configured to develop the reference voltage, V_{ref} that is provided to the non-inverting input of the OTA **338**.

A current (e.g., $X \cdot I$) that is a fraction of the current, I , flows through connection **316**. In an exemplary embodiment, X is less than one half or one fifth (e.g., X is 0.2, or 0.15, or less), and the current that flows through the connection **316** may be $0.1 \cdot I$.

In an exemplary embodiment, the resistances **317** and **319** form what is referred to as a resistive feedback structure using the fraction of the current, I , while maintaining a drain

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voltage at the transistor 326 as a reference voltage, V_{ref} , using a replica bias voltage, $V_{gate} = (R1/R1+R2) \cdot V_x$ (the voltage at node 315). The gate voltage, V_{gate} , appears at the node 318 and is also provided to the gate of the transistor 346 and to the gate of the transistor 374 through the resistance 376.

In an exemplary embodiment, the value of the resistance 317 may be $1.5 \times$ the value of the resistance 319.

The voltage $V_{n1} = V_{ref} + V_{gs1}$.

The voltage $V_{d1} = V_{n1} + V_{gs2}$.

The voltage V_{gs1} can be made to approximate the voltage V_{gs2} by sizing the transistor 334 as $(W1/L1)$ and the transistor 322 as $N1 \cdot (W1/L1)$, causing the voltage V_{d1} to approximate the voltage V_{ref} .

The voltage, V_{casc} , is generated using the voltage V_{ref} at node 325, thus making the drain voltage of the transistor 374 (the gain device of the amplifier) approximate to V_{ref} .

A current (e.g., $Y \cdot I$) that is a multiple of the current, I , flows through inductor 362 and connection 368. In an exemplary embodiment, Y is determined by a size of the transistor(s) 372 and/or 374 as compared to a size of other transistors in the amplifier circuit 300, for example the transistor(s) 342 and/or 346. In the example described above where the transistors 326, 346, and 342 have a size referred to as (W/L) , and the transistors 372 and 374 have a size that is referred to as $N \cdot (W/L)$, Y may equivalent to N such that the current that flows through the inductor 362 and/or the connection 368 is $N \cdot I$.

In an exemplary embodiment, the replica circuit 355 may be used to replace a second OTA that would otherwise be used to generate the voltage V_{ref} and the gate voltage, V_{gate} . In this manner, current consumption and circuit area are reduced.

In an exemplary embodiment, the current source 312, current source 328, and the components in the replica circuit 355 may form a first loop, or first current loop, configured to bias the transistor 374 without an OTA. In an exemplary embodiment, the OTA 338, transistor 342 and transistor 346 may form a second loop, or second current loop, configured to bias the transistor 372. In an exemplary embodiment, the current source 312, current source 328, and the components in the replica circuit 355 in the first current loop generate a first bias current for the amplifier core 360 that is proportional to a gain of the gain transistor, such as one or both of the transistors 346 and 374.

FIG. 4 is a graph 400 having a trace 410 showing the drain voltage, V_d , of an LNA gain transistor as a function of V_{ref} and temperature. The vertical axis 402 shows drain voltage, V_d , of a gain transistor, such as the transistor 374 of FIG. 3, and the horizontal axis 404 shows control voltage, V_{ref} , and temperature.

In an exemplary embodiment, the current, I_{mirror} , (FIG. 3) may have a proportional to absolute temperature (PTAT) characteristic, and the drain voltage, V_d , of the gain transistor, such as the transistor 374 of FIG. 3, may have a complementary to absolute temperature (CTAT) slope across temperature, which improves linearity at higher gain modes. The trace 410 shows the drain voltage, V_d , of the transistor 374 for different values of V_{ref} , across the variable resistance 337. The nomenclature “-30 25 85” refer to temperature in degrees Celsius. The trace 410 illustrates a CTAT behavior of the drain voltage, V_d , of the transistor 374, thus improving linearity at higher temperature.

FIG. 5 is a flow chart 500 describing an example of the operation of a method for biasing an amplifier. The blocks in

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the method 500 can be performed in or out of the order shown, and in some embodiments, can be performed at least in part in parallel.

In block 502, a bias voltage is generated. For example, the replica circuit 355 may generate a reference voltage, V_{ref} , to provide a reference for the OTA 338 and a voltage, V_{gate} , may be generated at node 318. The gate voltage, V_{gate} may be used to bias an amplifier.

In block 504, the bias voltage is applied to an amplifier. For example, the gate voltage, V_{gate} , is provided to the transistor 374 as a bias signal. The bias signal, V_{gate} , may be provided to the transistor 374 through a resistance 376, and may be used to bias the transistor 374. In an exemplary embodiment, a drain voltage, V_d , of the transistor 374, is dependent on the bias voltage, V_{gate} .

In block 506, a signal may be amplified. For example, the transistor 374 (to which the bias signal, V_{gate} , is applied) and/or the transistor 372 (e.g., a cascode transistor) may amplify a signal provided at connection 381.

FIG. 6 is a functional block diagram of an apparatus for biasing an amplifier (or other means for amplifying). The apparatus 600 comprises means 602 for generating a bias voltage. In certain embodiments, the means 602 for generating a bias voltage can be configured to perform one or more of the functions described in operation block 502 of method 500 (FIG. 5). In an exemplary embodiment, the means 602 for generating a bias voltage may comprise the replica circuit 355, for example configured to generate a reference voltage, V_{ref} , to provide a reference for the OTA 338 and generate a voltage, V_{gate} , at node 318. The gate voltage, V_{gate} may be used to bias an amplifier.

The apparatus 600 may also comprise means 604 for applying a bias voltage to an amplifier (or the other means for amplifying). In certain embodiments, the means 604 for applying a bias voltage to an amplifier can be configured to perform one or more of the functions described in operation block 504 of method 500 (FIG. 5). In an exemplary embodiment, the means 604 for applying a bias voltage to an amplifier may comprise the resistor 376, for example configured to provide the gate voltage, V_{gate} , to the transistor 374 as a bias signal.

The apparatus 600 may also include means 606 for amplifying or the amplifier referenced above in the description of the means 602 and 604. For example, the means for amplifying may include the transistor 374 (to which the bias signal, V_{gate} , is applied) and/or the transistor 372 (e.g., a cascode transistor).

Implementation examples are described in the following numbered clauses:

1. An amplifier circuit, comprising an amplifier core having a cascode transistor and a gain transistor; a bias circuit coupled to the amplifier core, the bias circuit comprising: a first current source; a second current source; an operational transconductance amplifier (OTA); a bias cascode transistor pair having a bias cascode transistor and a bias gain transistor; and a replica circuit coupled to the first current source and to the second current source, the replica circuit comprising: a first replica transistor coupled to the first current source, and a second replica transistor coupled to the second current source; a third replica transistor coupled to the second replica transistor; an adjustable resistance coupled to the first replica transistor; a voltage divider coupled to the second replica transistor and the third replica transistor, wherein a gate of the third replica transistor is configured to provide a bias voltage, V_{gate} , for the amplifier core.

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2. The amplifier circuit of clause 1, wherein the voltage divider comprises a first resistance and a second resistance, the gate of the third replica transistor coupled to a first node between the first resistance and the second resistance.

3. The amplifier circuit of clause 2, wherein the first resistance is coupled between the first node and a system ground.

4. The amplifier circuit of clause 2, wherein the second resistance is coupled between the first node and a drain of the second replica transistor.

5. The amplifier circuit of any of clauses 1 through 4, wherein the first replica transistor has a size $(W1/L1)$ and the second replica transistor has a size $N1*(W1/L1)$.

6. The amplifier circuit of any of clauses 1 through 5, wherein the third replica transistor has a size (W/L) .

7. The amplifier circuit of any of clauses 1 through 6, wherein a second node is formed between a source of the second replica transistor and a drain of the third replica transistor.

8. The amplifier circuit of clause 7, wherein the OTA has a non-inverting input coupled to the second node, an inverting input coupled to a third node formed between a source of the bias cascode transistor and a drain of the bias gain transistor.

9. The amplifier circuit of any of clauses 1 through 8, wherein an output of the OTA is a cascode bias signal applied to a gate of the bias cascode transistor and a gate of the cascode transistor.

10. The amplifier circuit of any of clauses 1 through 9, wherein the amplifier core is a low noise amplifier (LNA) core.

11. The amplifier circuit of any of clauses 1 through 10, wherein the first replica circuit comprises a first current loop; and the OTA comprises a second current loop.

12. The amplifier circuit of clause 11, wherein the first replica circuit is configured to generate a first bias current without an operational amplifier.

13. The amplifier circuit of clause 11, wherein the first current loop generates a first bias current for the amplifier core that is proportional to a gain of the gain transistor.

14. The amplifier circuit of any of clauses 1 through 13, wherein the voltage divider comprises a resistive feedback structure.

15. A method for biasing an amplifier, comprising: generating a bias voltage using a first current loop and a second current loop, the first current loop comprising a plurality of current sources and a voltage divider, the second current loop comprising a single amplifier; and applying the bias voltage to an amplifier.

16. The method of clause 15, wherein generating the bias voltage further comprises: generating a bandgap current; generating a mirror current; generating a fraction of the mirror current; and dividing the mirror current to provide the bias voltage at a first node.

17. The method of clause 16, further comprising generating a reference voltage at a second node, the reference voltage associated with the bandgap current.

18. The method of clause 17, further comprising generating a cascode voltage based on the reference voltage.

19. The method of any of clauses 17 or 18, wherein the reference voltage is applied to the single amplifier in the second current loop.

20. The method of any of clauses 15 through 19, wherein the amplifier is a low noise amplifier (LNA).

21. The method of clause 16, further comprising generating the bias voltage at the first node without an operational amplifier.

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22. The method of clause 20, wherein the bias voltage is proportional to a gain of a gain transistor in the LNA.

23. A device, comprising: means for generating a bias voltage using a first current loop and a second current loop, the first current loop comprising a plurality of current sources and a voltage divider, the second current loop comprising a single amplifier; and means for applying the bias voltage to an amplifier.

24. The device of clause 23, wherein the means for generating the bias voltage further comprises: means for generating a bandgap current; means for generating a mirror current; means for generating a fraction of the mirror current; and means for dividing the mirror current to provide the bias voltage at a first node.

25. The device of clause 24, further comprising means for generating a reference voltage at a second node, the reference voltage associated with the bandgap current.

26. The device of clause 25, further comprising means for generating a cascode voltage based on the reference voltage.

27. The device of any of clauses 23 through 26, wherein the amplifier is a low noise amplifier (LNA).

28. The device of clause 24, further comprising means for generating the bias voltage at the first node without an amplifier.

29. The device of clause 27, wherein the bias voltage is proportional to a gain of a gain transistor in the LNA.

The circuit architecture described herein described herein may be implemented on one or more ICs, analog ICs, RFICs, mixed-signal ICs, ASICs, printed circuit boards (PCBs), electronic devices, etc. The circuit architecture described herein may also be fabricated with various IC process technologies such as complementary metal oxide semiconductor (CMOS), N-channel MOS (NMOS), P-channel MOS (PMOS), bipolar junction transistor (BJT), bipolar-CMOS (BiCMOS), silicon germanium (SiGe), gallium arsenide (GaAs), heterojunction bipolar transistors (HBTs), high electron mobility transistors (HEMTs), silicon-on-insulator (SOI), etc.

An apparatus implementing the circuit described herein may be a stand-alone device or may be part of a larger device. A device may be (i) a stand-alone IC, (ii) a set of one or more ICs that may include memory ICs for storing data and/or instructions, (iii) an RFIC such as an RF receiver (RFR) or an RF transmitter/receiver (RTR), (iv) an ASIC such as a mobile station modem (MSM), (v) a module that may be embedded within other devices, (vi) a receiver, cellular phone, wireless device, handset, or mobile unit, (vii) etc.

Although selected aspects have been illustrated and described in detail, it will be understood that various substitutions and alterations may be made therein without departing from the spirit and scope of the present invention, as defined by the following claims.

What is claimed is:

1. An amplifier circuit, comprising:

an amplifier core having a cascode transistor and a gain transistor;

a bias circuit coupled to the amplifier core, the bias circuit comprising:

a first current source;

a second current source;

an operational transconductance amplifier (OTA);

a bias cascode transistor pair having a bias cascode transistor and a bias gain transistor; and

a replica circuit coupled to the first current source and to the second current source, the replica circuit comprising:

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- a first replica transistor coupled to the first current source, and a second replica transistor coupled to the second current source;
 - a third replica transistor coupled to the second replica transistor;
 - an adjustable resistance coupled to the first replica transistor;
 - a voltage divider coupled to the second replica transistor and the third replica transistor, wherein an output of the voltage divider is coupled to a gate of the third replica transistor to provide a bias voltage, V_{gate} , for the amplifier core.
2. The amplifier circuit of claim 1, wherein the voltage divider comprises a first resistance and a second resistance, the gate of the third replica transistor coupled to a first node between the first resistance and the second resistance.
 3. The amplifier circuit of claim 2, wherein the first resistance is coupled between the first node and a system ground.
 4. The amplifier circuit of claim 2, wherein the second resistance is coupled between the first node and a drain of the second replica transistor.
 5. The amplifier circuit of claim 1, wherein the first replica transistor has a size $(W1/L1)$ and the second replica transistor has a size $N1*(W1/L1)$.
 6. The amplifier circuit of claim 5, wherein the third replica transistor has a size (W/L) .

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7. The amplifier circuit of claim 1, wherein a second node is formed between a source of the second replica transistor and a drain of the third replica transistor.
8. The amplifier circuit of claim 7, wherein the OTA has a non-inverting input coupled to the second node, an inverting input coupled to a third node formed between a source of the bias cascode transistor and a drain of the bias gain transistor.
9. The amplifier circuit of claim 8, wherein an output of the OTA is a cascode bias signal applied to a gate of the bias cascode transistor and a gate of the cascode transistor.
10. The amplifier circuit of claim 1, wherein the amplifier core is a low noise amplifier (LNA) core.
11. The amplifier circuit of claim 1, wherein:
 - the replica circuit comprises a first current loop; and
 - the OTA comprises a second current loop.
12. The amplifier circuit of claim 11, wherein the replica circuit is configured to generate a first bias current without an operational amplifier.
13. The amplifier circuit of claim 11, wherein the first current loop generates a first bias current for the amplifier core that is proportional to a gain of the gain transistor.
14. The amplifier circuit of claim 1, wherein the voltage divider comprises a resistive feedback structure.

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